

Sarani Jayasooriya

Undergraduate Researcher — Computer Networking, Machine Learning & AI, UAVs, Cyber-Physical Systems

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SUMMARY

Award-winning undergraduate researcher specializing in **UAV-assisted networking, AI-driven network intelligence, and cyber-physical systems**. Recognized for **technical leadership** in national projects, fellowships, and competitive research programs. Current research focuses on **FANET routing, predictive link stability, and AI-assisted network optimization**.

EDUCATION

B.Sc. (Hons.) in Engineering — Computer Engineering

University of Peradeniya, Sri Lanka

CGPA: **3.83 / 4.00** (2022–Present)

Bachelor of ICT (Hons.) — Software Engineering

Asia e University, Malaysia

CGPA: **3.80 / 4.00** (2020–2024)

Pearson Assured Diploma — Human Resource Management

ESOFT Uni

High Distinction (2021–2022)

Pearson Assured Diploma — Digital Marketing

ESOFT Uni

High Distinction (2021–2022)

HONORS & FELLOWSHIPS

- NBQSA National ICT Awards 2025 — Merit Award (SkyT)
- Top 4 Finalist — SLASSCOM Xcellerate Inter-University Hackathon
- LKNOG9 Fellowship Recipient
- Winner — Inter-University Glider Competition 2024 (IESL)
- Top 25 Startup Founder — SLASSCOM × IITM Pravartak Startup Incubator Program, India

RESEARCH & TECHNICAL PROJECTS

Predictive Link Stability & Trust-Aware FANET Routing

Undergraduate Research Project (Ongoing)

- Studied link instability in Flying Ad-hoc Networks caused by high UAV mobility and dynamic topology changes.
- Proposed an AI-assisted routing framework incorporating mobility prediction, historical link quality, and trust metrics.
- Developed RL-based models to estimate link survival probability and route lifetime.
- Evaluated performance using simulation-based experiments with packet delivery ratio, latency, and route break frequency.

SkyT — Drone-Based Precision Agriculture

Cyber-Physical Systems Project (2024-2025)

- Designed an end-to-end cyber-physical platform integrating autonomous UAVs, IoT soil sensors, and cloud analytics.
- Implemented U-Net-based image segmentation for crop health assessment using aerial imagery.
- Established secure UAV–cloud communication and a web-based monitoring interface.
- Deployed scalable backend services using AWS, Cloudflare, MySQL, and MongoDB.
- Awarded Merit at the NBQSA National ICT Awards 2025.

BGP Traffic Analyzer using Recurrent Neural Networks

Pilot Project(2025)

- Developed a machine learning pipeline for traffic forecasting and anomaly detection in ISP-level BGP routing data.
- Applied sequence modeling to capture temporal dependencies and identify abnormal routing behavior.

CKDu Disease Progression Analysis using Machine Learning

Undergraduate Research Project(2023-2024)

- Designed a patient simulation model to study Chronic Kidney Disease of unknown etiology (CKDu).

- Developed predictive machine learning models using clinical indicators and longitudinal patient data.
- Implemented a full-stack web interface for visualization and exploratory analysis.

Agrovoltaic Systems for Tea Plantations

Research Project(Ongoing)

- Analyzed productivity and efficiency trade-offs in agrovoltaic tea plantation systems.
- Applied data analytics and machine learning techniques to support optimized land-use decisions.

MLink — Pharmacy Linking Platform

FullStack Development Project(2024)

- Developed a full-stack platform for real-time discovery of nearby medicine availability.
- Implemented API-based data retrieval, database integration, and microservice-oriented architecture.

EXPERIENCE

Undergraduate Teaching Assistant — University of Peradeniya

- Assisted Python programming and Computer Networking laboratory sessions.
- Evaluated laboratory submissions and supported protocol analysis and debugging.

Student Fellow — LKNOG9 Fellowship Program

- Collaborated with South Asian network engineers and researchers.
- Engaged in BGP security, IPv6 routing, ISP networks, and AI-driven anomaly detection.

TECHNICAL SKILLS

Programming Languages: Python, Java, C/C++, TypeScript, SQL

Networking: TCP/IP, BGP, SDN, OpenFlow, Mininet, GNS3

Machine Learning: CNNs, RNNs, U-Net, TensorFlow, PyTorch, Scikit-learn

Tools & Platforms: Linux, Git, Docker, Cloud Platforms

LEADERSHIP & VOLUNTEERING

Current Roles

- **Assistant Secretary — APAN 62 Conference** *2025 – Present*
 - Supported coordination of an international academic and networking conference involving global researchers and network operators.
 - Assisted in session planning, speaker coordination, and communication with technical and organizing committees.
- **Student Coordinator — ICPC Sri Lanka (University of Peradeniya)** *2025 – Present*
 - Designed and authored competitive programming problems for the ICPC Sri Lanka Regional Contest.
 - Ensured problem correctness, clarity, and appropriate difficulty through test case design and validation.
 - Collaborated with the organizing committee to align problems with ICPC standards and contest objectives.
 - Acted as a liaison between faculty coordinators and student teams.

Previous Roles

- **President — AeroGenZ Aviation Club, University of Peradeniya** *2022 – 2025*
 - Led a student aviation and UAV-focused club, organizing technical workshops, seminars, and hands-on projects.
 - Coordinated inter-university competitions and promoted student engagement in aerospace and drone technologies.
- **Junior Editor — IEEE Women in Engineering (WIE) Sri Lanka** *2023 – 2024*
 - Contributed to editorial activities including article review, content editing, and publication support.
 - Assisted in promoting technical outreach and visibility of women in engineering initiatives.

REFERENCES

Available upon request.